



SCIENCE AND TECHNOLOGY LEARNING OUTCOMES FOR PRIMARY SCHOOLS, BY CLASS LEVELS AND TEACHING CYCLES

STANDARD 2			
Cycle 1	Cycle 2	Cycle 3	Cycle 4
Introduction to Science and Technology /Scientific Method SC 1.01 Discuss who scientists are and what they do. SC 1.02 Outline the steps scientists use when conducting research. (Scientific Method) SC 1.03 Discuss and explain how science and technology are related and how they differ. Forces and Motion SC 2.01 Identify and describe force as a push or a pull that can change the speed and direction of an object. SC 2.02 Conduct and explain simple investigations related to gravity, friction and weight using appropriate tools such as ruler, scale, balance.	Earth Science SC 3.01 Discuss and explain how the earth's four spheres interact. SC 3.02 Identify and describe the components of the physical Earth as a system including the inner core, outer core, mantle, crust. SC 3.03 Identify and describe the components of the universe and give examples of how changes in one part of the universe affect other parts of the Earths-system. Hearing and Sound SC 4.01 Identify and differentiate between natural and manmade sounds. SC 4.02 Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.	Hearing and Sound SC 4.04 Demonstrate how the human ear detects sound and distinguish between the major components of the outer, middle and inner ear. Introduction to Digital Literacy/Information Literacy SC 5.01 Interact with and explore a range of digital equipment such as cameras, microphones, microscopes, laptops. SC 5.02 Describe how to position the body, adjust lighting, position equipment and when to take breaks when using digital devices. SC 5.03 Operate a variety of digital tools such as open/close, save, print, navigate, use input and output devices, log on and log off, and unplugging.	Classification of Living Things SC 6.01 Identify, define and classify animals as vertebrates and invertebrates. SC 6.02 Differentiate and explain the characteristics of living and non-living things and investigate the key differences between reptiles, amphibians, fishes, birds and mammals. SC 6.03 Observe, identify and classify plants and animals in Belize according to similarities and differences and attributes such as colour, size, sounds they make, smell, texture and behaviour. Parts of the Body and Senses SC 7.01 Characterize the structures associated with each of the 5 senses and make observations about materials and their environment using the sense organs.



STANDARD 2			
Cycle 1	Cycle 2	Cycle 3	Cycle 4
SC 2.03 Investigate, observe, and explain how magnets attract and repel certain objects.	SC 4.03 Compare frequency and pitch produced by different kinds of musical instruments.	SC 5.04 Identify, locate and use letters, numbers and special keys on keyboard such as space bar, shift and delete.	SC 7.02 Identify and name the main external parts of the body and their locations, such as the breast, vagina, penis, testicles and anus. SC 7.03 Examine the basic physical changes in people from birth to old age



STANDARD 3

Cycle 1	Cycle 2	Cycle 3	Cycle 4
Scientific Methods/Skills of a Scientist/ Tools of a Scientist SC 1.04 Conduct, investigate and solve problems in the homes and communities using the scientific skills and scientific method. SC 1.05 Plan and conduct scientific investigations using appropriate tools and techniques such as selecting an instrument that measures the desired quantity — length, volume, weight, time interval, temperature — with the appropriate level of precision. Simple Machines SC 2.04 Identify, demonstrate, and explain how a variety of simple machines work. SC 2.05 Observe, identify, and classify specific examples of simple machines. SC 2.06 Explore and utilize simple machines to make work easier and create	Rocks and Soils SC 3.05 Investigate and describe physical characteristics such as crystal size, shape, mineral and chemical composition, density and origin of igneous, metamorphic, and sedimentary rocks. SC 3.06 Identify the main types of soil commonly found in Belize and distinguish uses and characteristics of different types of soil. SC 3.07 Discuss and describe how soil forming factors may interact to produce a soil seen in different places in the landscape. Heat SC 4.05 Classify and illustrate natural and man-made sources of heat. SC 4.06 Investigate how adding or removing heat affects substances and their molecular state.	Heat SC 4.08 Demonstrate how heat can be transmitted through solids, liquids, and gases using conduction, convection, radiation. SC 4.09 Identify and investigate materials that are good heat insulators and good heat conductors and describe some uses of these materials. Online Safety/Surfing the Web SC 5.05 Identify the differences and Similarities between private and personal information and discuss why keeping personal information such as name, location, phone number and home addresses are considered private. SC 5.06 Discuss and explain the importance of password, passcode, and face Identification as a form of protection for private information. SC 5.07 Explain that devices Such as computers, laptops and tablets can save	Habitat, Animal Growth, and Development SC 6.06 Discuss and evaluate examples of environmental conditions that may threaten the survival of animals and plants that lives in Belize and other continent. SC 6.07 Determine, describe and investigate how selected plants and animals grow and discuss the factors that affect the rate of plant and animal growth. Body Parts, Organs, and Life Process SC 7.04 Identify major external parts of the human body, such as arms and hands; head and neck; legs and feet; ears, skin, eyes, nose and tongue and describe the function of each and how they relate to each other. SC 7.05 Name and identify major organs of the human body such as the brain, heart, lungs, stomach, liver, urinary



STANDARD 3			
Cycle 1	Cycle 2	Cycle 3	Cycle 4
an invention using one or more simple machine.	SC 4.07 Identify that heat energy is the most common by-product of energy transformations and describe some examples such as friction and body heat.	information as data that can be stored, searched, retrieved and deleted. Habitat, Animal Growth, and Development SC 6.04 Diagram typical growth cycles of local animals. SC 6.05 Show how a variety of local animals require different habitats during their growth cycle and discuss how habitats provide animals with their basic needs.	bladder, small intestine, large intestines, and kidneys. SC 7.06 Investigate how basic life processes such as simple respiration and digestion interact with each other in order to maintain healthy bodies



STANDARD 4			
Cycle 1	Cycle 2	Cycle 3	Cycle 4
Scientific Method/Branches of Science/Famous Local and International Scientist and their Contributions SC 1.06 Identify a national issue and use the scientific method to find a solution. SC 1.07 Obtain definitions of major branches of science and deduce where they are applied in everyday life. SC 1.08 Identify and connect famous local and international scientists to their contribution and their impact on everyday life. Floater and Flyers SC 2.07 Identify and compare features of some plants and animals that enables them to float. SC 2.08 Describe and demonstrate the relationship between the forces-lift,	Water for Life SC 3.08 Define the water cycle and explain the changes in state water undergoes at each step in the cycle. SC 3.09 Explain the importance of the water cycle to plants, animals, and human life and investigate how different human activities can impact the quality of water in Belize and other parts of the world. SC 3.10 List and discuss current efforts and technologies such as sonar technology, Satellite images, and Geographic information systems which are used to study, analyse and predict change in earth's water system.	Pure Substances and Mixtures SC 4.10 Define matter and explain the properties of solids, liquids, and gases based on the particle nature of matter. SC 4.11 Identify solutes and solvents in common solid, liquid, and gaseous solutions and discuss and differentiate between pure substances, mixtures, elements, and compounds by using the particle theory of matter. SC 4.12 Demonstrate different methods such as filtration, distillation and chromatography to separate the components of both solutions and mixtures. Cyberbullying, Online Etiquette (Netiquette) SC 5.08 Explain proper online etiquette.	Plant Growth and Development SC 6.08 Outline the main components of an ecosystem and create a representation of an aquatic or terrestrial ecosystem found in Belize. SC 6.09 Compare and contrast the roles of producers, consumers and decomposers in an aquatic and terrestrial food chain within the ecosystem. SC 6.10 Examine and distinguish between flowering and non-flowering plant species found locally and design and illustrate the growth cycles of flowering plants found locally Cells SC 7.07 Illustrate the relationship between the basic functions of cells, tissues, organs, and organ systems.



STANDARD 4			
Cycle 1	Cycle 2	Cycle 3	Cycle 4
weight, drag, and thrust that are required for flight. SC 2.09 Examine and explain why some objects sink and others float.		SC 5.09 Identify different forms of bullying, including cyberbullying and suggest strategies for dealing with it such as screenshot, saving evidence, block, not replying, report. SC 5.10 Identify and discuss online identity theft and security symbols such as padlock, phishing, scam websites.	SC 7.08 Describe the structure of a typical animal and plant cell and analyze their functions using models and other media.



STANDARD 5

Cycle 1	Cycle 2	Cycle 3	Cycle 4
Technology in Everyday Life SC 1.09 Explain how technology and climate change are interrelated. SC 1.10 Analyze and discuss what is climate change and how it affects Belize's tourism and agriculture sector. SC 1.11 Identify and describe how earth's weather patterns and conditions such as temperature and precipitation influence climate type and distribution and global regional perspective. Mechanical and Electrical system SC 2.10 Analyze and discuss examples of mechanical systems that are made up of a combination of simple machines. SC 2.11 Identify, demonstrate, and describe how the application of force can affect the movement of a mechanical device. SC 2.12 Build an electrical circuit with a battery, light bulb, and switch to determine and explain how it works.	Space Science SC 3.11 Describe and differentiate between stars, moons, planets, dwarf planets, meteors and asteroids of the solar system. SC 3.12 Research and describe current theories about the formation of the solar system, including the Nebular Theory. SC 3.13 Describe the Sun-Moon-Earth system and explain how revolution, rotation, and precession of the Sun-Moon-Earth system produce changes in the solar angle of incidence that result in seasons and changes in the length of a day, month, and year. SC 3.14 Describe current efforts and technologies used to study the universe, such as optical telescopes, radio telescopes, satellites, space probes, spectroscopes, and explain how their research impacts human activity.	Light and Optics SC 4.15 Demonstrate and explain that light is a form of energy, that travels in a straight line, and can be separated into the visible light spectrum. SC. 4.16 Conduct experiment to explore the differences between reflection and refraction of light and describe how reflected and refracted images formed. SC 4.17 Explain how the human eye detects images and compare the basic functional operation such as eye lids to shutter and lens to retina of the human eye to that of a camera in focussing an image. Ethical Use of Digital Resources SC 5.11 Define good digital citizenship as using technology safely, responsibly, and ethically.	Animal Diversity SC 6.11 Describe and explain the diversity of animals in ecosystems of Belize through characterization, mapping or artwork. SC 6.12 Discuss and illustrate the physical, behavioural and physiological adaptation features found in local animals. SC 6.13 Construct and use a dichotomous key to illustrate the classification of different local animal species. SC 6.14 Observe and describe seasonal migration of specific birds, insects and other animals in your local community. SC 6.15 Discuss ways to mitigate the impacts caused by humans on animal diversity in Belize.



STANDARD 5

Cycle 1

Cycle 2

Cycle 3

Cycle 4

Light and Optics

SC. 4.14 Identify a range of natural and man-made light sources and distinguish between objects that emit their own light such as sun, lightening, fire and those that require an external source of light such as moon, bicycle and car reflectors.

SC 5.12 Discuss that copying the work of others and presenting it as own is called plagiarism.

Human Systems

SC 7.09 Illustrate and explain the basic functions of the human respiratory, digestive, excretory, nervous, immune, skeletal and cardiovascular organ systems.

SC 7.10 Demonstrate the functions of one of the organ systems using role play, models digital or other media.



STANDARD 6

Cycle 1	Cycle 2	Cycle 3	Cycle 4
Benefits and Burdens of Science and Technology SC: 1.12 Discuss and debate the benefits and burdens of technology in the country of Belize. SC: 1.13 Explore what is a carbon footprint and deduce ways in which it affects climate change and describe practical strategies that can be implemented to reduce carbon footprints of an individual, a household or the country of Belize. SC: 1.14 Explore Global warming and deduce what strategies can be used to slow down this event. SC: 1.15 Explain how some countries are more responsible than others for producing greenhouse gas emissions and compare this with where the climate crisis has the most severe impacts.	Changes in an Ecosystem SC 3.15 Differentiate between and compare roles and components of Terrestrial and an Aquatic ecosystem. SC 3.16 Describe fresh and saltwater ecosystems in and around Belize and Investigate and explain how human activities may impact ecosystems. SC 3.17 Examine and explain the effects of the changes to a water body resulting from natural causes such as earthquakes, floods, storms, Tsunami and droughts in ecosystem Energy Resources SC 4.19 Define and classify energy resources into renewable and non-renewable resources and discuss if they are currently or potentially available in Belize.	Energy Resources SC 4.22 Analyze the immediate and long-term effects that extraction and uses of natural resources for energy production has on society and the environment and make recommendations for minimizing the effects now and in the future. Relationships and Communications Plagiarism SC 5.13 Identify and describe how students, teachers, parents, and other workers use many types of technologies in their daily work and personal lives. SC 5.14 Explain the similarities and differences between offline and online communications, including rules to follow when communicating face-to face and online and discuss how online communication can be misinterpreted. SC 5.15 List and explain the advantages of communicating electronically including time and resource saving, cost effectiveness, accessibility to multiple	Plant Diversity SC 6.16 Investigate and explain how selected plants have adapted in ways that enable them to survive in their ecosystem. SC 6.17 Examine and criticize how human activity has an impact on plant diversity and develop a plan to mitigate these impacts. SC 6.18 Justify how different natural changes to the ecosystem such as flooding, hurricanes and drought can affect the diversity of local plant species. Heredity and Human Reproduction SC 7.11 Explain and examine how the human male and female reproductive systems work with the aid of videos or websites.



STANDARD 6

Cycle 1	Cycle 2	Cycle 3	Cycle 4
Energy Conversions SC 2.13 Define, identify, and depict different types of energy transformations in daily life. SC 2.14 Design and conduct simple demonstrations to show that energy converts from one form to another.	SC 4.20 Investigate and create a model showing how hydropower, solar, and wind energy can be harnessed. SC 4.21 Describe and Justify how energy can be harnessed from different sources such as food, solar, wind, water, oil and gas, electrical, thermal, and wave for use by humans.	users from different countries communicating simultaneously, and ease of sharing, saving, or storing content. Plant Diversity SC 6.19 Investigate how selected plants have adapted in ways that enable them to survive in their ecosystem. SC 6.20 Explore diversity of plants in different ecosystems in Belize.	SC 7.12 Explore cells that contain DNA which passes on genetic information from one generation to the other. SC 7.13 Discuss and create a list of characteristics that can be inherited and those that cannot be inherited. SC 7.14 Investigate the use of selective reproduction by farmers and examine the effect on their animals and crops.